HOTEL RESERVATION & GUEST SERVICES MANAGEMENT SYSTEM

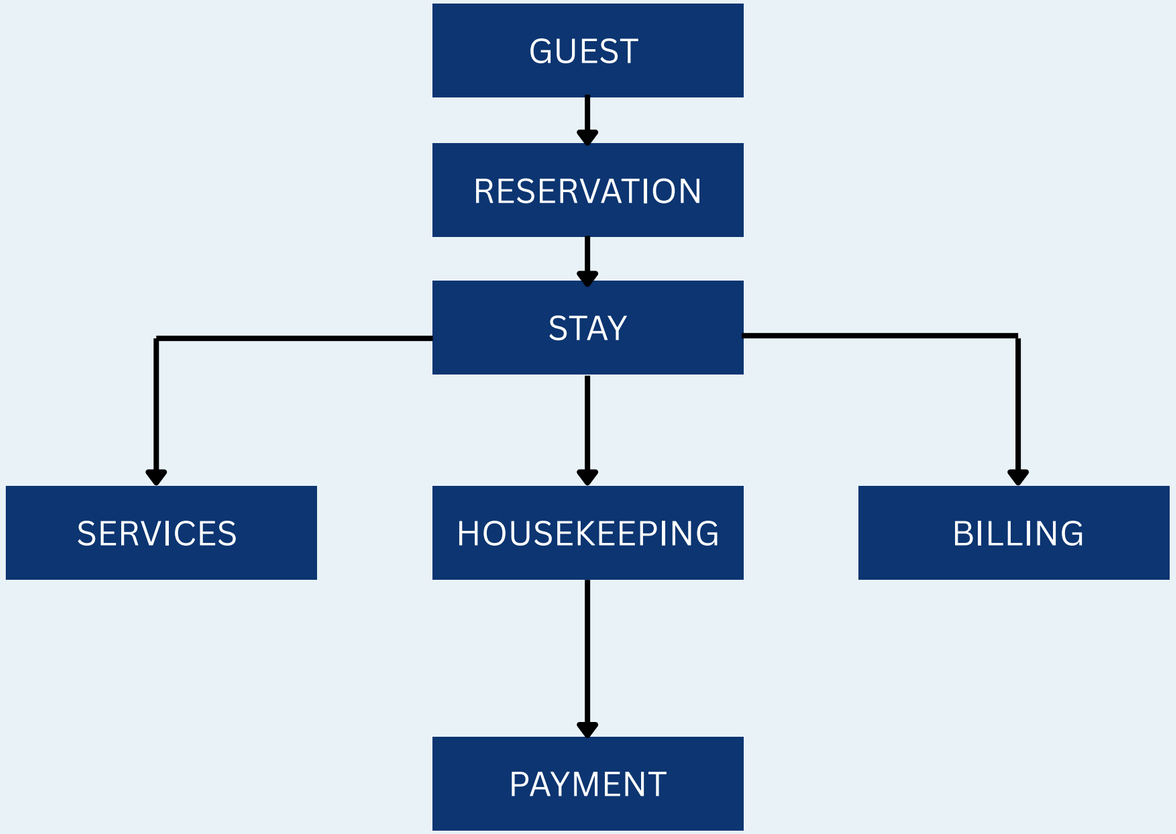
THE PROBLEM

Hotels manage multiple interconnected operations simultaneously.

- Room Availability
- Double Booking
- Guest Information
- Service Requests
- Housekeeping
- Billing

A CENTRALIZED HOTEL MANAGEMENT DATABASE

The proposed system maintains a single connected database for hotel operations.



OBJECTIVE

To design a centralized relational database system that efficiently manages hotel reservations, guest stays, services, housekeeping and financial transactions.

SYSTEM SCOPE

Room Management
- Room types
- Individual rooms
- Room capacity
- Room status

Guest Management
- Guest registration
- Contact information
- Guest history

Reservations & Stays
- Availability search
- Reservations
- Cancellations
- Check-in/out
- Stay tracking

Services & Housekeeping
- Service requests
- Service charges
- Housekeeping tasks
- Task assignment

Billing & Payments
- Invoices
- Room charges
- Service charges
- Payments

Reporting
- Occupancy
- Vacancies
- Cancellations
- Guest history
- Revenue

SYSTEM USERS

- Receptionist
- Housekeeping Staff
- Billing Staff
- Manager
- Guest

KEY RULES

01 — No Overlapping Stays

02 — Valid Dates

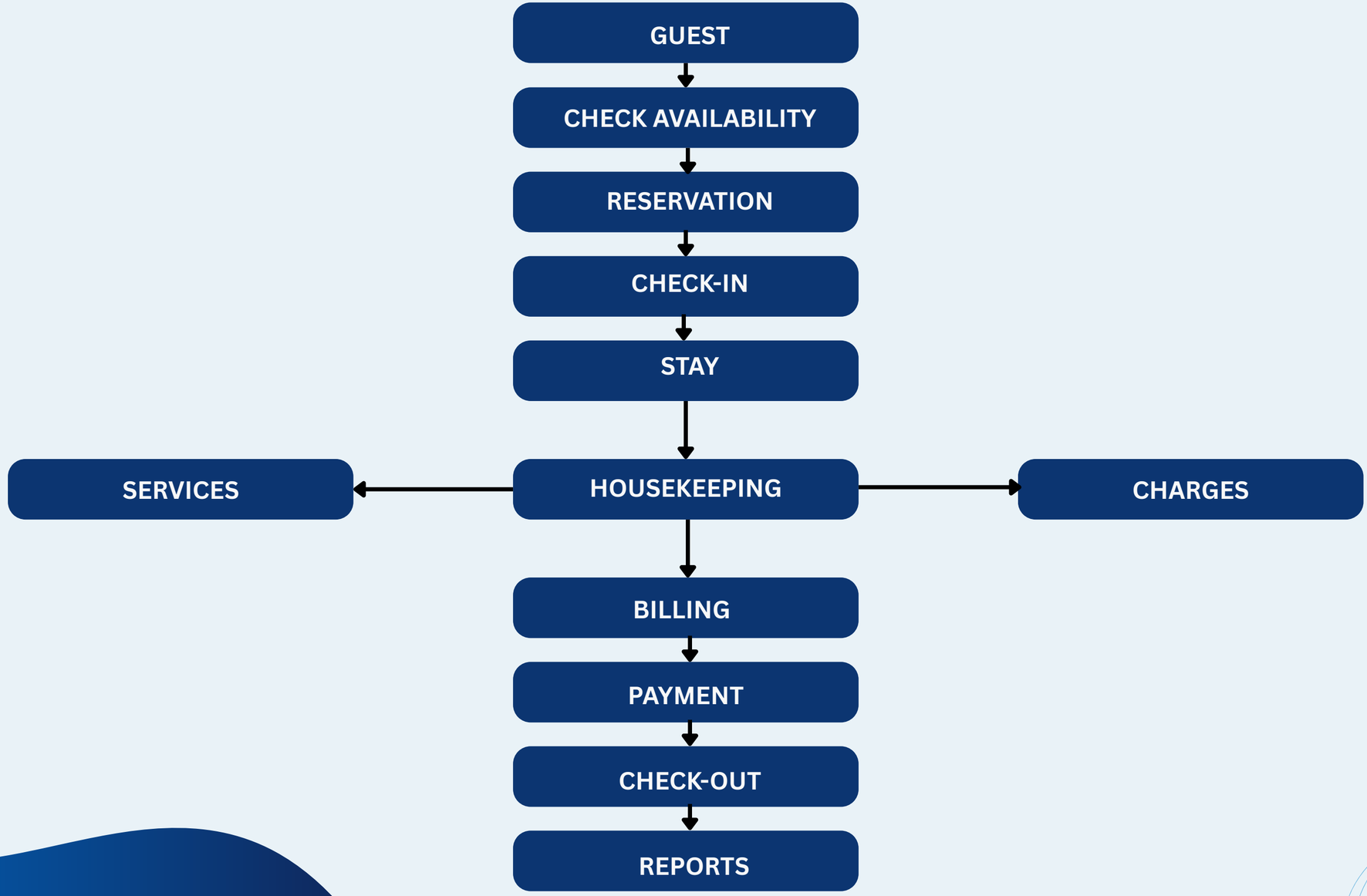
03 — Room Capacity

04 — Valid Charges

05 — Payment Limit

06 — Referential Integrity

SYSTEM WORKFLOW



ENTITIES & ATTRIBUTES

- ROOM_TYPE
- ROOM
- RATE_PLAN

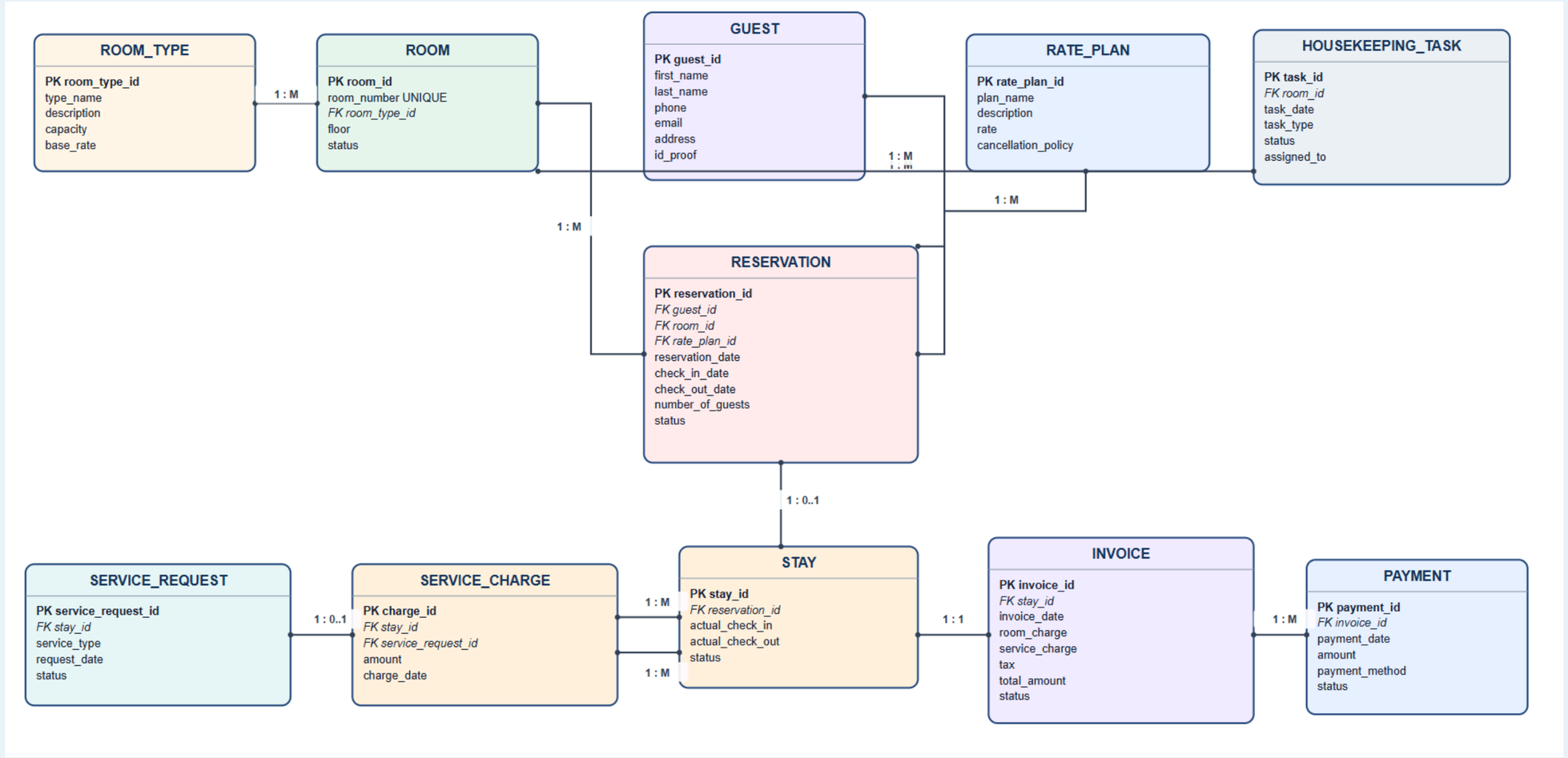
- GUEST
- RESERVATION
- STAY

- SERVICE_REQUEST
- HOUSEKEEPING_TASK
- SERVICE_CHARGE

- INVOICE
- PAYMENT

11 Principal Entities | 60+ Attributes | Multiple Interconnected Relationships

ER DIAGRAM



PK: Primary Key

FK: Foreign Key

INITIAL RELATIONAL SCHEMA

ROOM_TYPE
- (room_type_id PK, type_name, description, capacity, base_rate)

ROOM
- (room_id PK, room_number UNIQUE, room_type_id FK, floor, status)

RATE_PLAN
- (rate_plan_id PK, plan_name, description, rate, cancellation_policy)

GUEST
- (guest_id PK, first_name, last_name, phone, email, address, id_proof)

RESERVATION
- (reservation_id PK, guest_id FK, room_id FK, rate_plan_id FK, reservation_date, check_in_date, check_out_date, number_of_guests, status)

STAY
- (stay_id PK, reservation_id FK, actual_check_in, actual_check_out, status)

SERVICE_REQUEST
- (service_request_id PK, stay_id FK, service_type, request_date, status)

HOUSEKEEPING_TASK
- (task_id PK, room_id FK, task_date, task_type, status, assigned_to)

SERVICE_CHARGE
- (charge_id PK, stay_id FK, service_request_id FK, amount, charge_date)

INVOICE
- (invoice_id PK, stay_id FK, invoice_date, room_charge, service_charge, tax, total_amount, status)

PAYMENT

(payment_id PK, invoice_id FK, payment_date, amount, payment_method, status)

CONCLUSION

The proposed database provides a structured foundation for managing hotel reservations, guest stays, services, housekeeping and financial transactions. The conceptual design will serve as the basis for normalization and database implementation in the next phase.

NEXT PHASE:

Normalization → 3NF → DDL → Sample Data → SQL Queries & Reports